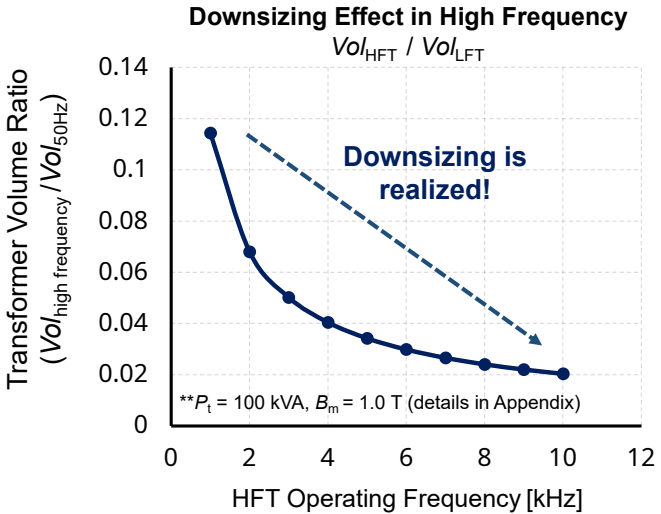
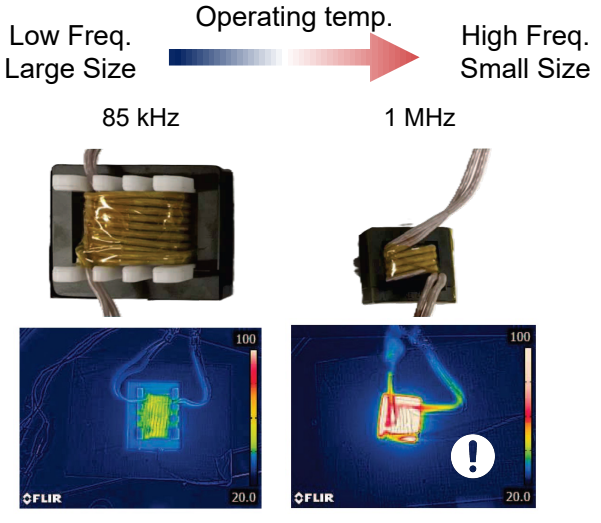

Downsizing Effect in High Frequency
Vol_{HFT} / Vol_{LFT}
Transformer Volume Ratio
($Vol_{high\ frequency}$/Vol_{50Hz})
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
Downsizing is realized!
**P_t = 100 kVA, B_m = 1.0 T (details in Appendix)
0
2
4
6
8
10
12
HFT Operating Frequency [kHz]

(a)


Low Freq.
Large Size
Operating temp.
High Freq.
Small Size
85 kHz
1 MHz
100
20.0
FLIR
100
20.0
FLIR

(b)